

```
import numpy as np
import pandas as pd
```

```
df= pd.read_csv('dataset_Facebook.csv', sep=';')
```

```
df.head()
```

	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour \
0	139441	Photo	2	12	4	
3						
1	139441	Status	2	12	3	
10						
2	139441	Photo	3	12	3	
3						
3	139441	Photo	2	12	2	
10						
4	139441	Photo	2	12	2	
3						

	Paid	Lifetime Post Total	Reach	Lifetime Post Total Impressions \
0	0.0		2752	5091
1	0.0		10460	19057
2	0.0		2413	4373
3	1.0		50128	87991
4	0.0		7244	13594

	Lifetime Engaged Users	Lifetime Post Consumers \
0	178	109
1	1457	1361
2	177	113
3	2211	790
4	671	410

	Lifetime Post Consumptions \
0	159
1	1674
2	154
3	1119
4	580

	Lifetime Post Impressions by people who have liked your Page \
0	3078
1	11710
2	2812
3	61027
4	6228

	Lifetime Post reach by people who like your Page \
0	1640

1	6112
2	1503
3	32048
4	3200

Lifetime People who have liked your Page and engaged with your post	
\	
0	119
1	1108
2	132
3	1386
4	396

	comment	like	share	Total Interactions
0	4	79.0	17.0	100
1	5	130.0	29.0	164
2	0	66.0	14.0	80
3	58	1572.0	147.0	1777
4	19	325.0	49.0	393

#create Subset

```
df1=df[['Page total likes', 'Category', 'Post Month', 'Post Weekday']].loc[0:15]
```

```
df2=df[['Page total likes', 'Category', 'Post Month', 'Post Weekday']].loc[16:30]
```

```
df3=df[['Page total likes', 'Category', 'Post Month', 'Post Weekday']].loc[31:50]
```

#Merging data

```
merging=pd.concat([df1,df2,df3])
merging
```

	Page total likes	Category	Post Month	Post Weekday
0	139441	2	12	4
1	139441	2	12	3
2	139441	3	12	3
3	139441	2	12	2
4	139441	2	12	2
5	139441	2	12	1
6	139441	3	12	1
7	139441	3	12	7
8	139441	2	12	7

9	139441	3	12	6
10	139441	2	12	5
11	139441	2	12	5
12	139441	2	12	5
13	139441	2	12	5
14	138414	2	12	4
15	138414	2	12	3
16	138414	3	12	3
17	138414	1	12	2
18	138414	3	12	2
19	138414	3	12	1
20	138414	2	12	1
21	138414	1	12	7
22	138414	1	12	7
23	138414	3	12	7
24	138414	2	12	6
25	138458	2	12	6
26	138458	2	12	5
27	138458	3	12	5
28	138895	2	12	5
29	138895	1	12	4
30	138895	2	12	4
31	138895	2	12	3
32	138895	3	12	3
33	138895	3	12	2
34	138895	1	12	2
35	138895	2	12	1
36	138895	3	12	1
37	138895	1	12	7
38	138895	2	12	7
39	138895	1	12	7
40	138895	2	12	6
41	138895	1	12	6
42	138353	1	12	5
43	138353	1	12	5
44	138353	1	12	4
45	138353	1	12	4
46	138353	1	12	3
47	138353	1	12	3
48	138353	1	12	2
49	138353	1	12	2
50	138353	2	11	1

#sort data

```
sort_values=df.sort_values('Page total likes', ascending=False)
sort_values
```

Page total likes	Type	Category	Post Month	Post Weekday
Post Hour \				

0		139441	Photo	2	12	4
3						
8		139441	Status	2	12	7
3						
1		139441	Status	2	12	3
10						
12		139441	Photo	2	12	5
10						
11		139441	Photo	2	12	5
10						
..		...	...	...	...	...
...						
495		85093	Photo	3	1	7
2						
496		81370	Photo	2	1	5
8						
497		81370	Photo	1	1	5
2						
498		81370	Photo	3	1	4
11						
499		81370	Photo	2	1	4
4						
	Paid	Lifetime Post Total	Reach	Lifetime Post Total	Impressions	
\						
0	0.0		2752		5091	
8	0.0		11844		22538	
1	0.0		10460		19057	
12	0.0		2847		5133	
11	0.0		3112		5590	
..	...		...		...	
495	0.0		4684		7536	
496	0.0		3480		6229	
497	0.0		3778		7216	
498	0.0		4156		7564	
499	NaN		4188		7292	
	Lifetime Engaged Users	Lifetime Post Consumers	\			
0	178	109				

8	1530	1407
1	1457	1361
12	193	115
11	208	127
..	...	...
495	733	708
496	537	508
497	625	572
498	626	574
499	564	524

Lifetime Post Consumptions \		
0	159	
8	1692	
1	1674	
12	133	
11	145	
..	...	
495	985	
496	687	
497	795	
498	832	
499	743	

Lifetime Post Impressions by people who have liked your Page \		
0	3078	
8	15220	
1	11710	
12	3779	
11	3887	
..	...	
495	4750	
496	3961	
497	4742	
498	4534	
499	3861	

Lifetime Post reach by people who like your Page \		
0	1640	
8	7912	
1	6112	
12	2072	
11	2174	
..	...	
495	2876	
496	2104	
497	2388	
498	2452	
499	2200	

```

Lifetime People who have liked your Page and engaged with your
post \
0 119
8 1250
1 1108
12 152
11 165
.. ...
495 392
496 301
497 363
498 370
499 316

```

```

comment like share Total Interactions
0 4 79.0 17.0 100
8 0 161.0 31.0 192
1 5 130.0 29.0 164
12 0 90.0 14.0 104
11 0 88.0 18.0 106
.. ...
495 5 53.0 26.0 84
496 0 53.0 22.0 75
497 4 93.0 18.0 115
498 7 91.0 38.0 136
499 0 91.0 28.0 119

```

```
[500 rows x 19 columns]
```

```
#Transpose data
```

```
transposing=df.transpose()
transposing
```

```

0 1
2 \
Page total likes 139441 139441
139441
Type Photo Status
Photo

```

Category	2	2
3		
Post Month	12	12
12		
Post Weekday	4	3
3		
Post Hour	3	10
3		
Paid	0.0	0.0
0.0		
Lifetime Post Total Reach	2752	10460
2413		
Lifetime Post Total Impressions	5091	19057
4373		
Lifetime Engaged Users	178	1457
177		
Lifetime Post Consumers	109	1361
113		
Lifetime Post Consumptions	159	1674
154		
Lifetime Post Impressions by people who have li...	3078	11710
2812		
Lifetime Post reach by people who like your Page	1640	6112
1503		
Lifetime People who have liked your Page and en...	119	1108
132		
comment	4	5
0		
like	79.0	130.0
66.0		
share	17.0	29.0
14.0		
Total Interactions	100	164
80		
	3	4
5 \		
Page total likes	139441	139441
139441		
Type	Photo	Photo
Status		
Category	2	2
2		
Post Month	12	12
12		
Post Weekday	2	2
1		
Post Hour	10	3
9		

Paid	1.0	0.0
0.0		
Lifetime Post Total Reach	50128	7244
10472		
Lifetime Post Total Impressions	87991	13594
20849		
Lifetime Engaged Users	2211	671
1191		
Lifetime Post Consumers	790	410
1073		
Lifetime Post Consumptions	1119	580
1389		
Lifetime Post Impressions by people who have li...	61027	6228
16034		
Lifetime Post reach by people who like your Page	32048	3200
7852		
Lifetime People who have liked your Page and en...	1386	396
1016		
comment	58	19
1		
like	1572.0	325.0
152.0		
share	147.0	49.0
33.0		
Total Interactions	1777	393
186		
	6	7
8 \		
Page total likes	139441	139441
139441		
Type	Photo	Photo
Status		
Category	3	3
2		
Post Month	12	12
12		
Post Weekday	1	7
7		
Post Hour	3	9
3		
Paid	1.0	1.0
0.0		
Lifetime Post Total Reach	11692	13720
11844		
Lifetime Post Total Impressions	19479	24137
22538		
Lifetime Engaged Users	481	537
1530		

Lifetime Post Consumers	265	232	
1407			
Lifetime Post Consumptions	364	305	
1692			
Lifetime Post Impressions by people who have li...	15432	19728	
15220			
Lifetime Post reach by people who like your Page	9328	11056	
7912			
Lifetime People who have liked your Page and en...	379	422	
1250			
comment	3	0	
0			
like	249.0	325.0	
161.0			
share	27.0	14.0	
31.0			
Total Interactions	279	339	
192			
	9	...	490
491 \			
Page total likes	139441	...	85979
85979			
Type	Photo	...	Photo
Photo			
Category	3	...	3
3			
Post Month	12	...	1
1			
Post Weekday	6	...	6
6			
Post Hour	10	...	11
3			
Paid	0.0	...	0.0
1.0			
Lifetime Post Total Reach	4694	...	5280
6184			
Lifetime Post Total Impressions	8668	...	8703
10228			
Lifetime Engaged Users	280	...	951
956			
Lifetime Post Consumers	183	...	911
901			
Lifetime Post Consumptions	250	...	1237
1140			
Lifetime Post Impressions by people who have li...	4309	...	5757
6085			
Lifetime Post reach by people who like your Page	2324	...	3300
3502			

Lifetime People who have liked your Page and en...	199	...	431
437			
comment	3	...	1
1			
like	113.0	...	79.0
105.0			
share	26.0	...	30.0
46.0			
Total Interactions	142	...	110
152			
	492	493	
494 \			
Page total likes	85979	85093	
85093			
Type	Link	Photo	
Photo			
Category	1	3	
3			
Post Month	1	1	
1			
Post Weekday	5	1	
7			
Post Hour	11	2	
10			
Paid	0.0	0.0	
0.0			
Lifetime Post Total Reach	45920	8412	
5400			
Lifetime Post Total Impressions	5808	13960	
9218			
Lifetime Engaged Users	753	1179	
810			
Lifetime Post Consumers	655	1111	
756			
Lifetime Post Consumptions	763	1632	
1003			
Lifetime Post Impressions by people who have li...	15766	8632	
5654			
Lifetime Post reach by people who like your Page	10720	5348	
3230			
Lifetime People who have liked your Page and en...	220	699	
422			
comment	0	17	
10			
like	128.0	185.0	
125.0			
share	9.0	55.0	
41.0			

Total Interactions	137	257
176		
	495	496
497 \		
Page total likes	85093	81370
81370		
Type	Photo	Photo
Photo		
Category	3	2
1		
Post Month	1	1
1		
Post Weekday	7	5
5		
Post Hour	2	8
2		
Paid	0.0	0.0
0.0		
Lifetime Post Total Reach	4684	3480
3778		
Lifetime Post Total Impressions	7536	6229
7216		
Lifetime Engaged Users	733	537
625		
Lifetime Post Consumers	708	508
572		
Lifetime Post Consumptions	985	687
795		
Lifetime Post Impressions by people who have li...	4750	3961
4742		
Lifetime Post reach by people who like your Page	2876	2104
2388		
Lifetime People who have liked your Page and en...	392	301
363		
comment	5	0
4		
like	53.0	53.0
93.0		
share	26.0	22.0
18.0		
Total Interactions	84	75
115		
	498	499
Page total likes	81370	81370
Type	Photo	Photo
Category	3	2
Post Month	1	1

Post Weekday	4	4
Post Hour	11	4
Paid	0.0	NaN
Lifetime Post Total Reach	4156	4188
Lifetime Post Total Impressions	7564	7292
Lifetime Engaged Users	626	564
Lifetime Post Consumers	574	524
Lifetime Post Consumptions	832	743
Lifetime Post Impressions by people who have li...	4534	3861
Lifetime Post reach by people who like your Page	2452	2200
Lifetime People who have liked your Page and en...	370	316
comment	7	0
like	91.0	91.0
share	38.0	28.0
Total Interactions	136	119

[19 rows x 500 columns]

#Shape and reshape

```
shaping=df.shape
shaping
```

(500, 19)

```
pivot_table=pd.pivot_table(df,index=['Type', 'Category'],
values='comment')
print(pivot_table)
```

Type	Category	comment
Link	1	2.900000
	2	2.000000
	3	2.000000
Photo	1	5.897297
	2	11.692308
	3	6.913333
Status	1	4.333333
	2	9.921053
	3	2.750000
Video	1	12.285714

```
reshaping_arr=np.array([1, 2, 3, 4, 5, 6])
reshaping_arr.reshape(3,2)
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```